

UNIT-V

Kinetics and Kinematics of Rigid Body part-4

1. Differentiate between kinetics and kinematics.
May, (2015)

Kinetics:

It deals motion of particles and bodies with consideration of forces.

Kinematics:

It deals motion of a body without consideration of forces.

2. Define impulse and momentum. (2015), May, May 2016,

Impulse:

It is defined as the product of force and the time

$$\text{Impulse (I)} = F \times t$$

Momentum:

It is defined as the product of mass and velocity

$$\text{Momentum (M)} = m v$$

m - mass of the particle

v - velocity of the particle.

3. What do you mean by general plane motion? (Jan, 2016)
(Dec. 2018)
General plane motion may be defined as the sum of translation and rotation.

e.g.:

4. Define instantaneous centre of rotation. (Jan, 2016)

The point which is instantaneously at rest and has zero velocity is called as the instantaneous centre of rotation.

5. Short notes on relative motion. (May, 2016)

When a motion of the body is evaluated with respect to another moving body, then the motion is said to be relative motion.

6. Find angular acceleration of a body, if its angular velocity is increased from 8 rad/s to 28 rad/s in 5 sec. (May 2017)

Solution:

Given:

$$\omega_0 = 8 \text{ rad/sec.}$$

$$\omega = 28 \text{ rad/sec.}$$

$$t = 5 \text{ sec.}$$

Angular acceleration (α) =

$$\omega = \omega_0 + \alpha t$$

$$28 = 8 + \alpha(5)$$

$$5\alpha = 28 - 8 = 20$$

$$\alpha = \frac{20}{5} = \underline{\underline{4 \text{ rad/sec}^2}}$$

7. State the principle of work and energy applicable for a rigid body (May 2017), (Nov 2016)

It states that $K.E_1 + W = K.E_2$,
where

$K.E_1$ - The body's initial translational and rotational kinetic energy

$K.E_2$ - The body's final translational and rotational kinetic energy.

W - work done by the external forces.

8. Define work and energy. (Dec 2017)

Work:

It is the product of the force and displacement along the direction of force.

$$\text{work done, } W = F \times S.$$

Energy:

The capacity of doing work is known as energy.

9. Define absolute motion. (Nov. 2016)

It is defined as the change of position of a body from one absolute place to another

10. Write an equation of principle of impulse and momentum (Dec. 2018)

$$(\Sigma F) t = m(v - u)$$

t — time in sec.

m — mass in kg

v — Final velocity in m/sec

u — Initial velocity in m/sec.

part B

1. An automobile of 1500 kg is being driven down 8° incline at a speed of 80 km/h. When the brakes are applied causing a braking force of 7 kN, to be applied to the automobile. Determine the time required for the automobile to come to rest. Use the impulse and momentum principle and take the coefficient of friction between the road and the tyre as 0.4 (May 2015)

Solution:

Given

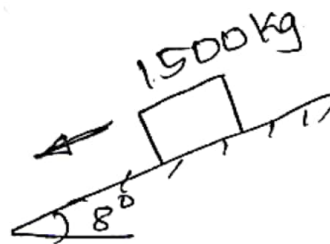
$$m = 1500 \text{ kg}$$

$$\text{Braking force } P = 7 \text{ kN} = 7000 \text{ N}$$

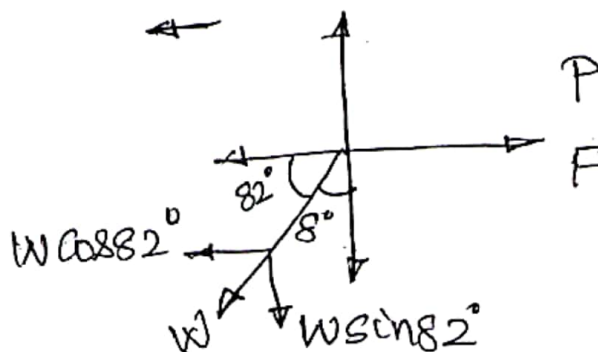
$$v = 0$$

$$\mu = 0.4$$

$$u = \frac{80 \times 10^3}{60 \times 60} \\ = 22.22 \text{ m/sec}$$



Free Body



$$W = mg \\ = 1500 \times 9.81 \\ = 14715 \text{ N}$$

$$F = \mu R$$

$$= 0.4 \times 14715$$

$$= \underline{5886 \text{ N}}$$

Applying Impulse Momentum Equation,

$$(\sum F) t = m(v - u)$$

$$(W \cos 82^\circ - P - F) t = 1500(0 - 22.22)$$

$$(14715 \cos 82^\circ - 1000 - 5886) t = -1500 \times 22.22$$

$$(2047.93 - 1000 - 5886) t = -33,330$$

$$+ 10,838.07 t = -33,330$$

$$t = \frac{33,330}{10,838.07}$$

$$= \underline{3 \text{ sec}}$$

Time required for the automobile to come to rest, $t = \underline{3 \text{ sec}}$

2. A body of weight 1 kN is on the horizontal surface of the table. This weight is connected another body of weight 2 kN by a string passing over a smooth pulley fixed at the corner of the table. The coefficient of friction between 1 kN weight and the table surface is 0.20. If the system is released from the rest, find the velocity of the 2 kN weight after it has moved 1.2 m using the work-energy method. (May 2015)

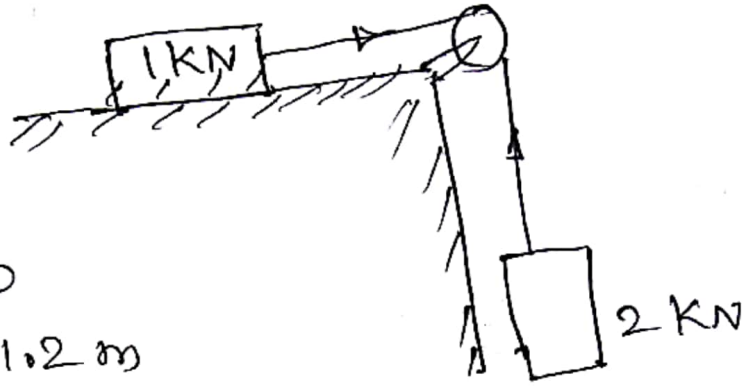
Solution:

Given:

$$u = 0$$

$$S = 1.2 \text{ m}$$

$$\mu = 0.20$$



Consider 1 kN weight:

Work Energy Equation,

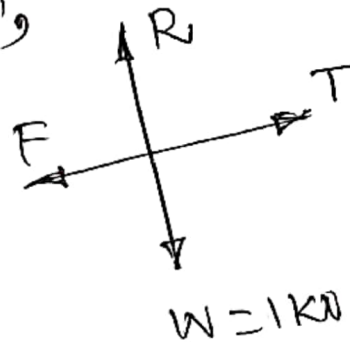
FBD

→ Direction of motion of weight

$$(\Sigma F) S = \frac{W}{2g} (V^2 - u^2)$$

$$(T - F) S = \frac{1000}{2 \times 9.81} (V^2 - 0)$$

$$(T - 200) 1.2 = \frac{1000 V^2}{2 \times 9.81}$$



$$\begin{aligned} \Sigma F_y &= 0 \\ R - W &= 0 \\ R &= 1 \text{ kN} \\ &= 1000 \text{ N} \\ F &= \mu R \\ &= 0.2 \times 1000 \\ &= \underline{200 \text{ N}} \end{aligned}$$

$$T - 200 = 42.473 V^2 \quad \text{--- (1)}$$

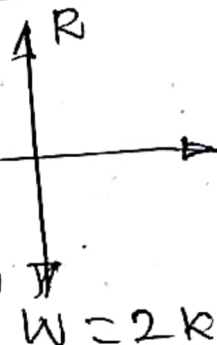
Consider 2 kN weight:

$$(\Sigma F) S = \frac{W}{2g} (V^2 - u^2)$$

$$(-T + 2000) 1.2 = \frac{2000 (V^2 - 0)}{2 \times 9.81}$$

$$-T + 2000 = 84.947 V^2 \quad \text{--- (2)}$$

FBD



$$\begin{aligned} \textcircled{1} + \textcircled{2} &\Rightarrow \cancel{x} - 200 = 42.473v^2 \\ &\quad -\cancel{x} + 2000 = 84.947v^2 \\ \hline 1800 &= 127.42v^2 \end{aligned}$$

$$v^2 = \frac{1800}{127.42} = 14.126$$

$$\therefore v = 3.75 \text{ m/sec}$$

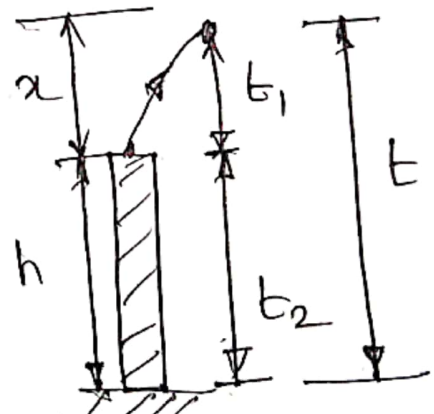
3. A stone is projected vertically upwards from the top of a building with a velocity of 35 m/s. Five seconds later another stone is just released from the same point. It is seen that both the stones reach the ground at the same time. Find the height of the building. (May 2016)

Solution:

Given:

$$u = 35 \text{ m/sec}$$

$$g = 9.81 \text{ m/sec}^2$$



t = total time taken by the stone to strike the ground.

t_1 = time taken to reach maximum height

t_2 = time taken to strike the ground,

$$t = t_1 + t_2$$

Upward Motion:

$$v = u - gt,$$

$$0 = 35 - 9.81t,$$

$$9.81t_1 = 35$$

$$t_1 = \frac{35}{9.81} = 3.56 \text{ sec}$$

$$\underline{\underline{= 3.6 \text{ sec}}}$$

$$x = ut_1 - \frac{1}{2}gt_1^2$$

$$= 35(3.6) - \frac{1}{2} \times 9.81(3.6)^2$$

$$= 126 - 17.658 = 63.57$$

$$= 108$$

$$\underline{\underline{= 62.43 \text{ m}}}$$

Downward motion:

$$h + x = ut_2 + \frac{1}{2}gt_2^2$$

$$= 0 + \frac{1}{2} \times 9.81(t - t_1)^2$$

$$= \frac{1}{2} \times 9.81(t - 3.6)^2$$

$$h + 62.43 = 4.9(t - 3.6)^2 \quad \text{--- (1)}$$

Motion of the 2nd stone:

$$u = 0,$$

$$t = (t - 5)$$

$$h = ut + \frac{1}{2}gt^2$$

$$= 0 + \frac{1}{2} \times 9.81(t - 5)^2$$

$$h = 4.9(t - 5)^2 \quad \text{--- (2)}$$

① - ②,

$$h + 62.43 = 4.9(t - 3.6)^2$$

$$h = 4.9(t - 5)^2$$

$$62.43 = 4.9[(t - 3.6)^2 - (t - 5)^2]$$

$$= 4.9[t^2 - 7.2t + 12.96 - (t^2 - 10t + 25)]$$

$$= 4.9[t^2 - 7.2t + 12.96 - t^2 + 10t - 25]$$

$$= 4.9[2.8t - 12.04]$$

$$12.74 = 2.8t - 12.04$$

$$2.8t = 12.74 + 12.04$$

$$= 24.78$$

$$t = \frac{24.78}{2.8}$$

$$= \underline{\underline{8.85 \text{ Sec}}}$$

$$= \underline{\underline{9.85 \text{ Sec}}}$$

② $\Rightarrow$

$$h = 4.9(t - 5)^2$$

$$= 4.9(8.85 - 5)$$

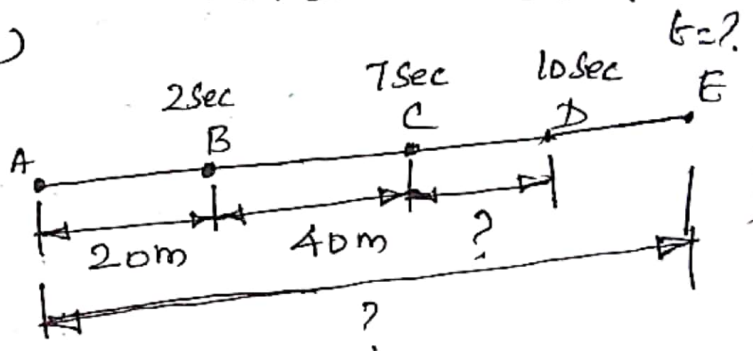
$$= 4.9(3.85)$$

$$= \underline{\underline{18.865 \text{ m}}}$$

$\therefore$ Height of the building, $h = \underline{\underline{18.865 \text{ m}}}$

7. A particle under constant deceleration is moving in a straight line and covers a distance of 20 meters in the first 2 seconds and 40 meters in the next 5 seconds. Calculate the distance it covers in the subsequent 3 seconds and the total distance travelled by the particle before it comes to rest.
- Solution: (May 2016)

Let the particle starts from A and come to halt at E as shown in fig.



Consider the motion between A and B,

$$s = ut + \frac{1}{2}at^2$$

$$20 = u(2) + \frac{1}{2}a(2)^2$$

$$20 = 2u + 2a$$

$\div$ by 2 $\Rightarrow$

$$10 = u + a \quad \text{--- (1)}$$

consider the motion between A and C,

$$60 = 7u + \frac{1}{2}a(7)^2$$

$$60 = 7u + 24.5a \quad \text{--- (2)}$$

$$\textcircled{1} \times 7 \Rightarrow 70 = 7u + 7a \quad \text{--- (3)}$$

$$\textcircled{2} - \textcircled{3} \Rightarrow 60 = 7u + 24.5a$$

$$70 = 7u + 7a$$

$$-10 = 17.5a$$

$$a = -0.571 \text{ m/sec}^2$$

$$= -0.571 \text{ m/sec}^2$$

Sub in ①,

$$10 = u + a$$

$$u = 10 - a \\ = 10 - (-0.571) = 10.571 \text{ m/sec.}$$

Let distance AD be equal to S_1 ,

$$S_1 = 10.571 \times 10 + \frac{1}{2}(-0.571)(10)^2 \\ = \underline{77.16 \text{ m}}$$

$\therefore$ distance covered in the interval, 7 sec. to 10 sec.,

$$CD = 77.16 - 60 = \underline{17.16 \text{ m}}$$

Let the particle come to rest at a distance S from the starting point i.e. $AE = S$,

$$AE = S$$

$$v^2 - u^2 = 2aS$$

$$0 - (10.571)^2 = 2(-0.571)S$$

$$S = \frac{(10.571)^2}{2 \times 0.571} = \underline{97.85 \text{ m}}$$

2. power supply was cut-off to a power driven wheel when it was rotating at a speed of 900 rpm. It was observed to come to rest after making 360 revolutions. Determine its angular retardation and time it took to come to rest after power supply was cut-off. (NOV. 2016)

Solution:

Given:

$$N_0 = 900 \text{ rpm}$$

$$\omega = 0$$

$$\omega_0 = \frac{2\pi N_0}{60} \\ = \frac{2\pi \times 900}{60}$$

$$\omega_0 = 30\pi \text{ rad/sec}$$

$$\theta = 2\pi N$$

$$\theta = 2\pi \times 360$$

$$= 720\pi \text{ rad}$$

$$\omega^2 - \omega_0^2 = 2\alpha\theta$$

$$(0)^2 - (30\pi)^2 = 720\pi \times 2\alpha$$

$$\alpha = -1.9635 \text{ rad/sec}^2$$

$$\omega = \omega_0 + \alpha t$$

$$0 = 30\pi - 1.9635t$$

$$1.9635t = 30\pi$$

$$t = \frac{30\pi}{1.9635} = \underline{\underline{48 \text{ sec}}}$$

6. The angular acceleration of a flywheel is given by $\alpha = 12 - t$, where α is in rad/sec^2 and t is in sec. If the angular velocity of the flywheel is 60 rad/sec at the end of 4 sec, determine angular velocity at the end of 6 sec. How many revolutions take place in these 6 sec?

(NOV. 2016)

Solution:

$$\alpha = 12 - t$$

$$\frac{d\omega}{dt} = 12 - t$$

$$d\omega = (12 - t) dt$$

$$\int d\omega = \int (12 - t) dt$$

Integ,

$$\omega = 12t - \frac{t^2}{2} + C_1 \quad \text{--- (1)}$$

When $t = 4 \text{ sec}$

$$\omega = 12(4) - \frac{(4)^2}{2} + C_1,$$

$$\left| \omega = 60 \text{ rad/sec} \right.$$

$$60 = 12(4) - \frac{4^2}{2} + C_1,$$

$$C_1 = 20$$

Sub in (1),

$$\omega = 12t - \frac{t^2}{2} + 20$$

When $t = 6 \text{ sec}$,

$$\omega = 12(6) - \frac{(6)^2}{2} + 20$$

$$= 74 \text{ rad/sec}$$

$$\omega = \frac{d\theta}{dt}$$

$$\omega dt = d\theta$$

$$(12t - \frac{t^2}{2} + 20) dt = d\theta$$

Integ,

$$\int (12t - \frac{t^2}{2} + 20) dt = \int d\theta$$

$$\theta = 6t^2 - \frac{t^3}{6} + 20t + C_2$$

When $t = 0$, $C_2 = \theta_0$

When $t = 6 \text{ sec}$,

$$\theta_6 = 6(6)^2 - \frac{(6)^3}{6} + 20(6) + C_2$$

$$= 180 + C_2$$

Angular displacement during 6 sec, $= \theta_6 - \theta_0$

$$= 180 + C_2 - C_2$$

$$= 180 \text{ rad}$$

$$\text{NO. of rev} = \frac{180}{2\pi} = \underline{\underline{28.648}}$$

7. Two bodies weighing 300 N and 450 N are hung to the ends of a rope passing over an ideal pulley. What would be acceleration of heavier body, if it moves in downward direction? What is the tension developed in the rope? Using the principle of work and energy, obtain the solution. (May 2019)

Solution:

Block 300 N:

FBD

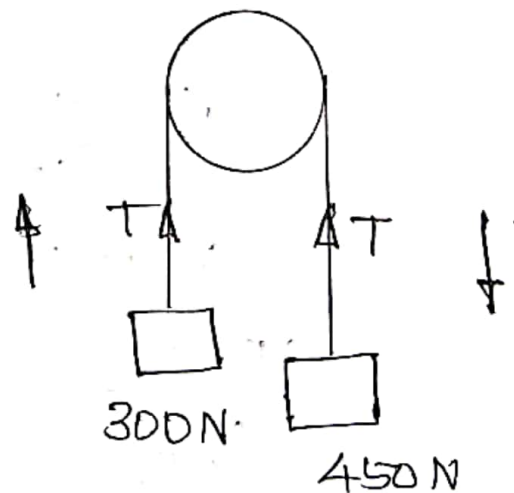


$$W = 300 \text{ N}$$

Work Energy equation,

$$(\Sigma F) s = \frac{W}{2g} (v^2 - u^2)$$

$$(T - 300)s = \frac{300}{2 \times 9.81} 2as$$



$$v^2 - u^2 = 2as$$

$$T - 300 = 600a$$

$$T - 300 = \frac{2 \times 9.81}{30.581a} \quad \text{--- (1)}$$

Block 450 N:

FBD



$$(\Sigma F)s = \frac{W}{2g}(v^2 - u^2)$$

$$(-T + 450)s = \frac{450}{2 \times 9.81} 2a$$

$$W = 450 \text{ N}$$

$$-T + 450 = \frac{900a}{2 \times 9.81}$$

$$-T + 450 = 45.87a \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow \cancel{T} - 300 = 30.58a$$

$$\cancel{-T} + 450 = 45.87a$$

$$150 = 76.45a$$

$$a = \frac{150}{76.45} = 1.96 \text{ m/sec}^2$$

sub in (1),

$$T - 300 = 30.581(1.96)$$

$$T - 300 = 60$$

$$T = 60 + 300 = \underline{\underline{360 \text{ N}}}$$

Tension in the rope, $T = \underline{\underline{360 \text{ N}}}$

8. The system of connected bodies is shown in Fig. Determine the velocity of Block A, 10 seconds after release from rest and the tension in the cable. Block A and B weigh 10 kg and 15 kg respectively. The coefficient of friction between the Block A and the surface is 0.25. The angle of inclination of the plane is 30° . (May 2018)

Solution:

Given:

$$t = 10 \text{ sec}$$

$$u = 0$$

$$\mu = 0.25$$

$$m_A = 10 \text{ kg}$$

$$m_B = 15 \text{ kg}$$

Block A:

FBD

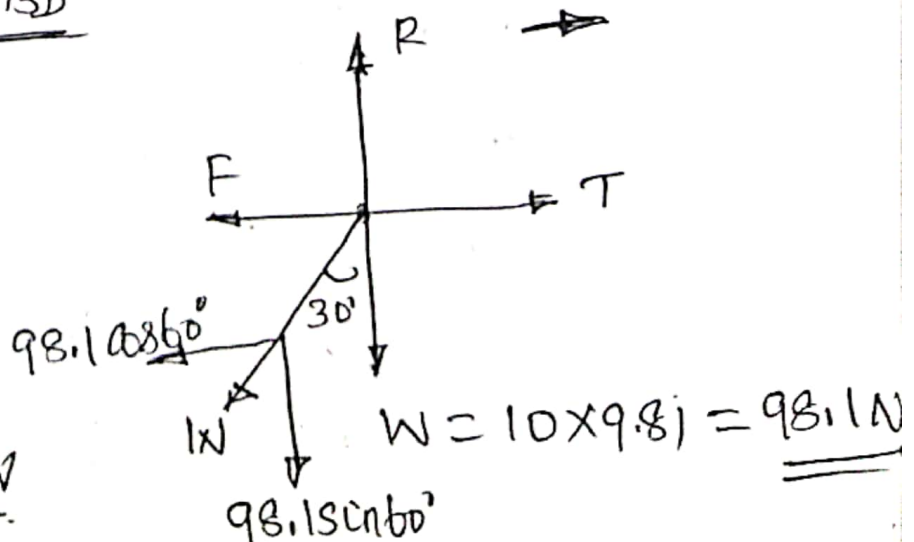
$$\sum F_y = 0$$

$$R - W \sin 60^\circ$$

$$R = W \sin 60^\circ$$

$$= 98.1 \sin 60^\circ$$

$$= \underline{84.957 \text{ N}}$$



Impulse Momentum eqn,

$$(\Sigma F) t = m(v - u)$$

$$F = \mu R$$

$$= 0.25 \times 84.957$$

$$= \underline{21.239 \text{ N}}$$

$$(T - F - 98.1 \cos 60^\circ) 10 = 10(v - 0)$$

$$(T - 21.239 - 49.05) 10 = 10v$$

$$T - 70.289 = v \quad \text{--- (1)}$$

Block B:

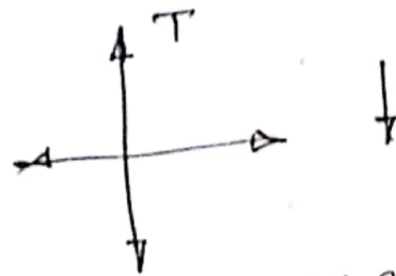
FBD

$$(\Sigma F) t = m(v - u)$$

$$(-T + W) 10 = 15(v - 0)$$

$$(-T + 147.15) 10 = 15v$$

$$-T + 147.15 = 1.5v \quad \text{--- (2)}$$



$$W = 15 \times 9.81$$

$$= 147.15 \text{ N}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow T - 70.289 = v$$

$$-T + 147.15 = 1.5v$$

$$\underline{76.861 = 2.5v}$$

$$v = \frac{76.861}{2.5} = \underline{\underline{30.74 \text{ m/sec}}}$$

Sub in (1),

$$T - 70.289 = v$$

$$T = v + 70.289$$

$$= 30.74 + 70.289 = \underline{\underline{101 \text{ N}}}$$

9. A flywheel weighing 50 kN having radius of gyration 1 m losses its speed from 400 rpm to 280 rpm in 2 minutes. Calculate:

- retarding torque acting on it,
- change in kinetic energy during the above period and
- change in angular momentum during the same period. (May 2017)

$$\omega_0 = \frac{2\pi N_0}{60} = \frac{2\pi \times 400}{60} = 41.888 \text{ rad/sec.}$$

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 280}{60} = 29.322 \text{ rad/sec.}$$

$$t = 2 \text{ min} = 2 \times 60 = 120 \text{ sec}$$

$$\omega = \omega_0 + \alpha t$$

$$29.322 = 41.888 + \alpha(120)$$

$$\text{retardation, } \alpha = -0.1047 \text{ rad/sec}^2$$

$$\text{weight of the flywheel} = 50 \text{ kN} \\ = 50 \times 10^3 \text{ N}$$

$$\text{Radius of gyration, } k = 1 \text{ m}$$

$$I = mk^2$$

$$= \frac{W}{g} k^2$$

$$= \frac{50000}{9.81} \times 1^2 = 5096.84$$

$$1) \text{ Retarding torque acting on the flywheel} = I\alpha$$

$$I\alpha = 5096.84 \times 0.1047$$

$$= 533.74 \text{ N}\cdot\text{m}$$

ii) Change in Kinetic Energy = Initial K.E - Final K.E

$$= \frac{1}{2} I \omega_0^2 - \frac{1}{2} I \omega^2$$

$$= \frac{I}{2} (\omega_0^2 - \omega^2)$$

$$= \frac{1}{2} \times 5096.84 (41.888^2 - 27.322^2)$$

$$= 22,804,429 \text{ N}\cdot\text{m}$$

iii) change in its angular momentum = $I\omega_0 - I\omega$

$$= I(\omega_0 - \omega)$$

$$= 5096.84 (41.888 - 27.322)$$

$$= 64048.93 \text{ N}\cdot\text{sec}$$

1a. The rotation of a flywheel is governed by the equation $\omega = 3t^2 - 2t + 2$ where ω is in radians per second and t is in sec. After one second from the start, the angular displacement was 4 radians, determine the angular displacement, angular velocity and angular acceleration of the flywheel when $t = 3$ seconds. (Sep. 2020)

Solution:

$$\omega = 3t^2 - 2t + 2$$

$$\omega = \frac{d\theta}{dt}$$

$$\omega dt = d\theta$$

$$\int \omega dt = \int d\theta$$

$$\int (3t^2 - 2t + 2) dt = \int d\theta$$

$$\theta = \frac{3t^3}{3} + \frac{-2t^2}{2} + 2t + C$$

$$\theta = t^3 - t^2 + 2t + C$$

$$t = 1; \theta = 4 \text{ rad.}$$

$$4 = 1 - 1 + 2(1) + C$$

$$4 = 2 + C$$

$$C = 2$$

$$\theta = t^3 - t^2 + 2t + 2$$

$$t = 3$$

$$\theta = (3)^3 - 3^2 + 2(3) + 2$$

$$\underline{\theta = 26 \text{ rad.}}$$

$$\omega = \frac{d\theta}{dt} = \frac{d}{dt}(3t^2 - 2t + 2) = 6t - 2$$

$$\omega dt = d\theta$$

$$\int \omega dt = \int d\theta \quad | \quad \omega \neq$$

$$t = 3 \text{ sec.}$$

$$\omega = 6(3) - 2 = \underline{\underline{16 \text{ rad/sec}^2}}$$

$$\theta = 3t^2 - 2t + 2$$

$$= 3(3)^2 - 2(3) + 2$$

$$= 27 - 6 + 2 = \underline{\underline{23 \text{ rad/sec}}}$$